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Cu/N-Oxyl-catalyzed aerobic oxidative esterification to oxalic acid diesters from ethylene glycol *via* highly selective intermolecular alcohol oxidation†

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One of the ideal green esterification reactions is aerobic oxidative esterification using only a stoichiometric amount of different alcohols *via* intermolecular selective alcohol oxidation followed by hemiacetal formation by the addition of the other alcohol and hemiacetal oxidation to esters. However, oxalic acid diester synthesis *via* oxidative esterification has not been reported to date, possibly owing to the difficulty of selectivity control of intermolecular alcohol oxidation and the chelating effects of ethylene glycol-derived alcohols/hemiacetals on inhibiting oxidation catalysts. Herein, using a CuCl/tetramethylethylenediamine/1,5-dimethyl-9-azanoradamantane *N*-oxyl catalyst, we describe a highly efficient aerobic oxidative esterification reaction of ethylene glycol to various oxalic acid diesters *via* selective oxidation of ethylene glycol-derived alcohols/hemiacetals even in the presence of other aliphatic primary alcohols. Notably, the green reaction works well using an ideal stoichiometric ratio of ethylene glycol and primary/secondary alcohols. Thorough experimental investigation and theoretical calculations revealed that highly selective oxidative esterification is enabled by the preferential bidentate coordination of ethylene glycol-derived alcohols/hemiacetals to the Cu(II) species, followed by efficient two-electron/one-proton transfer.

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Introduction

Esterification of alcohols, which are functional motifs ubiquitously found in a wide range of compounds, is a fundamental and important reaction to synthesize esters.¹ One of the ideal green esterification reactions is aerobic oxidative esterification using only a stoichiometric amount of two kinds of alcohols *via* selective oxidation of one primary alcohol without oxidizing the other alcohol (Fig. 1I). The oxidative esterification reaction proceeds *via* selective aerobic oxidation of an alcohol to an aldehyde (step 1, Fig. 1I-a), followed by nucleophilic addition of the other alcohol to the aldehyde to form a hemiacetal intermediate (step 2, Fig. 1I-a) and subsequent selective aerobic oxidation of the hemiacetal to the desired ester (step 3, Fig. 1I-a).² Differently from well-known reversible esterification processes such as Fischer esterification³ and transesterification,⁴ oxidative esterification involves an irreversible oxidation reaction,² which theoretically enables the use of stoichiometric

amounts of alcohols to complete the esterification. Unlike typical ester synthesis from carboxylic halides, esterification using O₂ as a terminal oxidant would be environmentally friendly, as only H₂O would be produced as a byproduct. However, most of the previously reported oxidative esterification reactions using different alcohols are limited to methyl esterification with methanol and/or esterification of benzyl alcohols (Fig. 1I-b).^{5–12} In addition, almost all of the esterification reactions require excessive amounts of nucleophilic alcohols (Fig. 1I-b).^{6–14} To our knowledge, oxidative esterification using a stoichiometric ratio of different alcohols is extremely rare and limited to the use of benzyl alcohol to date,⁵ mainly because the selective intermolecular oxidation of different alcohols of the same class is quite difficult, especially for aliphatic primary alcohols, based on numerous previously reported alcohol oxidation reactions.^{15–27}

Oxalic acid diesters are valuable compounds used as chemiluminescent reagents,²⁸ raw materials for pharmaceutical synthesis,^{29–31} catalytic ligands³² and substrates of coupling reactions.^{33,34} Despite the expected impact that an environmentally friendly and versatile method for the synthesis of oxalic acid diesters *via* oxidative esterification of ethylene glycol, a common and inexpensive feedstock,³⁵ would have on the synthetic chemistry field, such an approach has not been

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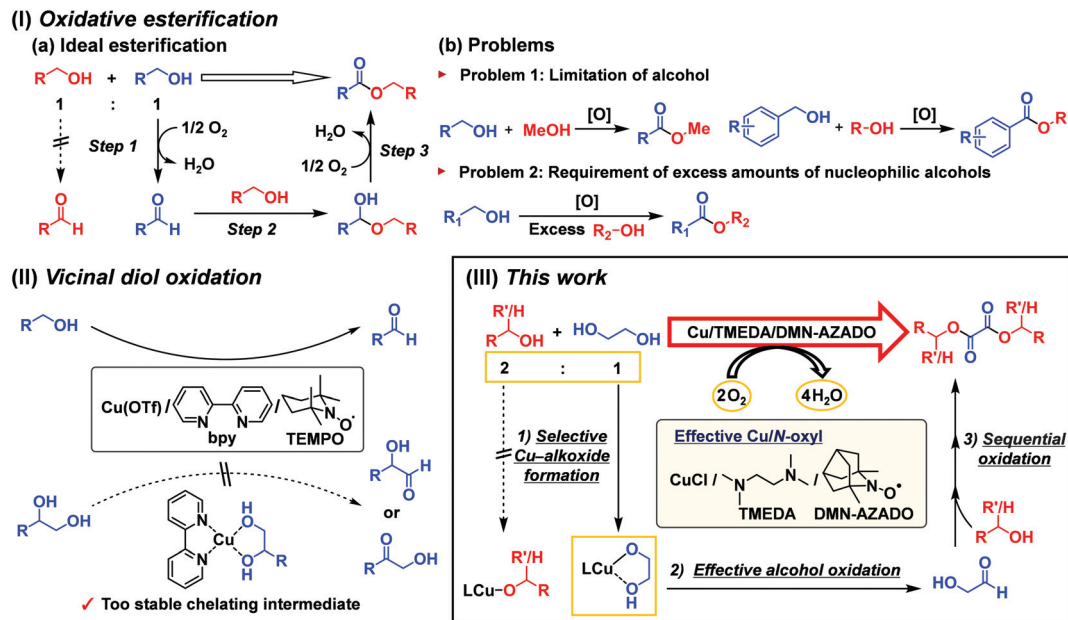


Fig. 1 Key problems and established system for the selective aerobic oxidative esterification at a stoichiometric ratio of different alcohols. (I) oxidative esterification of different alcohols: (a) reaction pathway of ideal oxidative esterification; (b) problems of conventional oxidative esterification. (II) Difficulty in the oxidation of vicinal diols using Cu/N-oxyl catalysts. (III) This work: highly selective aerobic oxidative esterification from a stoichiometric amount of ethylene glycol and various alcohols to oxalic acid diesters using CuCl/TMEDA/DMN-AZADO as a catalyst.

reported to date.^{36–39} In addition to the difficulty of selectivity control of intermolecular alcohol oxidation, this is possibly because the chelation of vicinal diols like ethylene glycol⁴⁰ has been reported to cause the inhibition of catalytic reactions. Stahl *et al.* reported that although a combined catalyst of Cu species and aminoxyl radicals (Cu/N-oxyl),^{41–47} Cu(i) triflate (Cu(OTf))/2,2'-bipyridyl (bpy)/2,2,6,6-tetramethylpiperidine N-oxyl (TEMPO), can promote the oxidation of various alcohols and diols, it hardly oxidizes vicinal diols owing to too stable complex formation *via* the chelating effects (Fig. 1II).^{46,48} Therefore, this ethylene glycol-based oxidative esterification seems to require a highly active alcohol oxidation catalyst working even in the presence of chelating effects. Recently, we reported an imide synthesis from alcohols and amides *via* oxidative acylation catalyzed by CuCl/N,N,N',N'-tetramethylethylenediamine (TMEDA)/9-azanoradamantane N-oxyl (nor-AZADO).⁴⁹ We also found that, unlike the aforementioned Cu(OTf)/bpy/TEMPO catalyst, the CuCl/TMEDA/nor-AZADO system catalyzed the oxidation of a few vicinal diols, including styrene glycol and 1,2-hexanediol.⁴⁹ On the basis of this result, we envisioned that an unprecedented oxidative esterification at a stoichiometric ratio to oxalic acid diesters could be achieved in the presence of a further tuned Cu/N-oxyl catalyst by utilizing chelating effects as a strategy for selective intermolecular alcohol oxidation.

Herein, we report the successful development of an aerobic oxidative esterification reaction of ethylene glycol to various oxalic acid diesters *via* selective oxidation of ethylene glycol-derived alcohols/hemiacetals in the presence of different alcohols, including aliphatic primary alcohols, using CuCl/TMEDA/

1,5-dimethyl-9-azanoradamantane N-oxyl (DMN-AZADO) as a catalyst (Fig. 1III). Surprisingly, the environmentally friendly esterification to oxalic acid diesters proceeded very well even with ideal stoichiometric amounts of ethylene glycol and alcohols. Various experiments and theoretical calculations revealed the origin of the high selectivity and activity for the aerobic oxidation of ethylene glycol-derived alcohols/hemiacetals.

Results and discussion

Optimization of reaction conditions

Initially, we performed the oxidative esterification of ethylene glycol (**1b**) with ethanol (**1a**) under 1 atm of O₂ at room temperature (~23 °C) in benzotrifluoride. We started our investigation by examining the CuCl/TMEDA/nor-AZADO catalyst system previously used for the oxidative acylation of amides with alcohols.⁴⁹ However, using CuCl (5 mol%)/TMEDA (5 mol%)/nor-AZADO (3 mol%) and activated molecular sieves (MS-3A, 200 mg) afforded the desired diethyl oxalate (**1c**) in 30% yield after 10 min (Table 1, entry 1). Then, other N-oxyls such as 9-azabicyclo [3.3.1] nonane N-oxyl (ABNO), 9-azabicyclo [3.3.1] nonane-3-one N-oxyl (keto-ABNO), 2-azaadamantane N-oxyl (AZADO), 1-methyl-2-azaadamantane N-oxyl (1-Me-AZADO), DMN-AZADO and TEMPO were also examined for the oxidative esterification of **1b** with **1a** (Table 1, entries 2–7). As a result, all N-oxyls except for TEMPO with CuCl/TMEDA afforded **1c** selectively. Cu/N-oxyl catalysts based on less sterically hindered N-oxyls including ABNO have been previously reported to show high activity for the oxidation of primary

Table 1 Effect of *N*-oxyl on the synthesis of **1c** from **1a** and **1b**

$2 \text{ 1a} + \text{1b} \xrightarrow[\text{O}_2]{\text{CuCl/TMEDA/N-oxyl}} \text{1c} + \text{1d}$

Entry	<i>N</i> -oxyl	Conversion (%)		Yield (%)	
		1a	1b	1c	1d
1	nor-AZADO	63	>99	30	3
2	ABNO	65	>99	32	2
3	keto-ABNO	67	>99	36	3
4	AZADO	92	>99	64	7
5	1-Me-AZADO	92	>99	65	8
6	DMN-AZADO	97	>99	84	5
7	TEMPO	14	40	<1	3

nor-AZADO

ABNO

keto-ABNO

AZADO

1-Me-AZADO

DMN-AZADO

TEMPO

Reaction conditions: **1a** (1.1 mmol), **1b** (0.55 mmol), CuCl (0.025 mmol), TMEDA (0.025 mmol), *N*-oxyl (0.015 mmol), benzonitrile (1.6 mL), O₂ (balloon), MS-3A (200 mg), room temperature, 10 min, chlorobenzene (PhCl) (0.1 mmol) as an internal standard. Conversions and yields were determined using gas chromatography analysis. For a detailed discussion on mass balances, see the ESI (Fig. S1 and Table S1).†

alcohols and hemiacetals,⁵⁰ and ABNO has proved to be highly effective for the oxidative lactonization of non-vicinal diols.⁴⁷ However, in the present study, only low yields of **1c** were obtained using ABNO or keto-ABNO (Table 1, entries 2 and 3). In contrast, when using AZADO or 1-Me-AZADO, which have more sterically hindered structures than nor-AZADO, **1c** was obtained in relatively high yields (Table 1, entries 4 and 5). Finally, CuCl/TMEDA and DMN-AZADO, which is more sterically hindered than 1-Me-AZADO, exhibited the highest catalytic performance for oxidative esterification, producing **1c** in 84% yield (Table 1, entry 6). It should be noted that the reaction hardly proceeded with TEMPO, probably owing to its excessive bulkiness (Table 1, entry 7).

In general, less sterically hindered *N*-oxyls mediate effectively the oxidation of hemiacetals because hemiacetals are more sterically hindered than primary alcohols.⁴⁷ Thus, it was surprising that DMN-AZADO, which is a comparatively sterically hindered *N*-oxyl, was more effective for the oxidative esterification of **1b** with **1a**. To gain a mechanistic insight into the unique role of DMN-AZADO, we compared the reactivities of DMN-AZADO and nor-AZADO. We initially evaluated the reaction profiles in the synthesis of **1c** from **1a** (4.4 mmol) and **1b** (2.2 mmol) with CuCl/TMEDA (0.025 mmol) and the respective *N*-oxyls (0.015 mmol) (Fig. 2a). At the initial stage of the reaction, nor-AZADO was more effective than DMN-AZADO. However, after a few minutes, the catalytic activity of CuCl/nor-AZADO decreased. Finally, the yield of **1c** became higher in the case of DMN-AZADO, suggesting that the steric hindrance of *N*-oxyls affected the catalytic lifetimes in this reaction, as we previously reported.⁵¹ In the Cu/*N*-oxyl-catalyzed oxidation of alcohols using a less sterically hindered *N*-oxyl such as

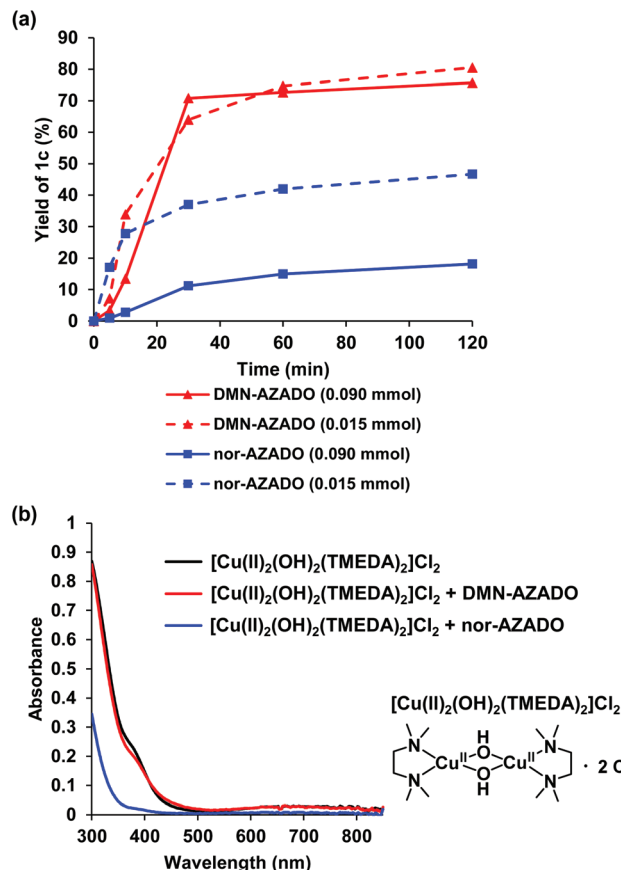


Fig. 2 Comparison of DMN-AZADO and nor-AZADO: (a) reaction profiles of the synthesis of **1c** from **1a** and **1b**; (b) UV-vis absorption spectra of Cu/*N*-oxyl complexes. Reaction conditions: (a) **1a** (4.4 mmol), **1b** (2.2 mmol), CuCl (0.025 mmol), TMEDA (0.025 mmol), *N*-oxyl (0.015 mmol or 0.090 mmol), benzonitrile (6.4 mL), O₂ (balloon), MS-3A (800 mg), room temperature, PhCl (0.4 mmol) as an internal standard. The values of DMN-AZADO (0.090 mmol) are the average of two runs. Yields were determined using gas chromatography analysis. Measurement conditions: (b) CH₃CN solution (1.5 mL) containing [Cu(II)₂(OH)₂(TMEDA)₂]Cl₂ (0.0125 mmol), *N*-oxyl (0.015 mmol) and MS-3A (200 mg), stirring for 1 h at room temperature under an O₂ atmosphere.

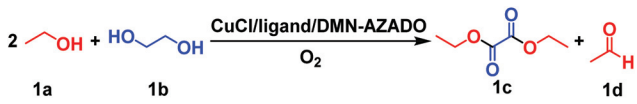
AZADO, a hydroxyl amine generated from the *N*-oxyl reacts with aldehydes to produce esters *via* oxidative esterification.⁵² In contrast, DMN-AZADO has two methyl groups around its radical site, and the reaction of the corresponding hydroxyl amine with aldehydes is hindered by steric effects, possibly prolonging the catalytic lifetimes. Then, to confirm whether other *N*-oxyl effects are present in this reaction, we conducted the reaction using an increased amount of *N*-oxyls (0.090 mmol). Surprisingly, increasing the amount of nor-AZADO inhibited the synthesis of **1c**, whereas an increased amount of DMN-AZADO promoted the reaction (Fig. 2a).

To verify the reasons for this inhibition, Cu/*N*-oxyl complexes were characterized by ultraviolet-visible (UV-vis) absorption spectroscopy and cold-spray ionization mass spectrometry (CSI-MS, positive mode) (Fig. 2b and S2, ESI†). The UV-vis absorption spectrum of a Cu(II)/TMEDA complex, [Cu(II)₂(OH)₂(TMEDA)₂]Cl₂, exhibited a ligand-to-metal charge

transfer band of bridging O(H) to Cu(II) at 355 nm.⁵³ When DMN-AZADO was added to the Cu(II) complex, this absorption decreased slightly but did not disappear. In contrast, when nor-AZADO was added to the solution, the absorption at 355 nm almost disappeared, suggesting that nor-AZADO coordinated to the Cu(II) center more easily than DMN-AZADO. Moreover, the CSI-MS spectrum of a CH₃CN solution of CuCl, TMEDA and DMN-AZADO presented a weak set of signals at $m/z = 361.26$ assignable to a DMN-AZADO-coordinated Cu(II)/TMEDA species, and the peaks derived from CuCl/TMEDA were retained (Fig. S2, ESI†). Meanwhile, the CSI-MS spectrum of a CH₃CN solution of CuCl, TMEDA and nor-AZADO exhibited a strong set of signals at $m/z = 333.23$, which can be ascribed to a species of nor-AZADO coordinated to Cu(II)/TMEDA, and almost no peaks derived from CuCl/TMEDA were observed (Fig. S2, ESI†). These CSI-MS results are consistent with the aforementioned UV-vis results. Therefore, less sterically hindered *N*-oxyls like nor-AZADO seem to coordinate easily to the Cu(II) center, possibly inhibiting the alcohol coordination and suppressing the oxidative esterification. From the above, it can be concluded that CuCl/DMN-AZADO exhibited the appropriate steric hindrance to suppress side reactions with aldehydes and inhibition of alcohol coordination to the Cu(II) center, thereby affording high catalytic performance.

Other commonly utilized ligands^{46,47,49,51,54} such as *N,N,N',N'*-tetramethyldiaminomethane (TMMDA), *N,N,N',N'*-tetramethyl-1,3-diaminopropane (TMPDA), *N,N,N',N'*-tetramethyl-1,4-diaminobutane (TMBDA), *N,N,N',N'*-tetraethylethylenediamine (TEEDA), bpy, 4,4'-dimethyl-2,2'-bipyridyl (^{4Me}bpy), 6,6'-dimethyl-2,2'-bipyridyl (^{6Me}bpy), 4,4'-dimethoxy-2,2'-bipyridyl (^{4MeO}bpy), 1,10-phenanthroline (phen), pyridine, 4-dimethylaminopyridine (DMAP) and quinuclidine were also examined for the oxidative esterification of **1b** with **1a** (Table 2 and Fig. S3, ESI†). However, these ligands were not effective, and **1c** was hardly obtained under the present reaction conditions (Table 2, entries 2–13); that is, only TMEDA was able to promote this reaction effectively (Table 2, entry 1). This unique TMEDA effect can be explained to some extent. On the one hand, TMEDA forms a stable five-membered-ring Cu complex, whereas an unstable four-membered-ring Cu complex is produced by TMMDA (Table 2, entry 2). TMPDA, TMBDA and TEEDA, which have longer methylene chains than TMEDA, form larger Cu complexes, possibly inhibiting the coordination of alcohols and *N*-oxyls to the complexes owing to steric effects (Table 2, entries 3–5). On the other hand, aromatic bidentate nitrogen ligands such as bpy, ^{4Me}bpy, ^{6Me}bpy, ^{4MeO}bpy and phen, which form five-membered rings like TMEDA, were much less effective, probably because they were electron-deficient compared with TMEDA (Table 2, entries 6–10). As with our previous work,⁴⁹ TMEDA possibly accelerates the reoxidation of Cu(I) species effectively owing to its electron-rich characteristics. Moreover, we confirmed by careful gas chromatography (GC) and GC-MS analyses that TMEDA was almost preserved under the present oxidation conditions, and byproducts derived from TMEDA were not detected at all after the reaction.

Table 2 Effect of the ligand on the synthesis of **1c** from **1a** and **1b**

					
Entry	Ligand	Conversion (%)		Yield (%)	
		1a	1b	1c	1d
1	TMEDA	97	>99	84	5
2	TMMDA	16	60	<1	3
3	TMPDA	18	68	<1	1
4	TMBDA	25	54	<1	1
5	TEEDA	20	89	2	<1
6	bpy	21	47	<1	1
7	^{4Me} bpy	25	58	<1	2
8	^{6Me} bpy	20	41	<1	<1
9	^{4MeO} bpy	20	50	<1	<1
10	Phen	18	59	<1	<1
11 ^a	Pyridine	30	66	1	10
12 ^a	DMAP	44	78	4	13
13 ^a	Quinuclidine	12	26	<1	1

Reaction conditions: **1a** (1.1 mmol), **1b** (0.55 mmol), CuCl (0.025 mmol), ligand (0.025 mmol), DMN-AZADO (0.015 mmol), benzotrifluoride (1.6 mL), O₂ (balloon), MS-3A (200 mg), room temperature, 10 min, PhCl (0.1 mmol) as an internal standard. Conversion and yields were determined using gas chromatography analysis. ^a Ligand (0.05 mmol). For a detailed discussion on mass balances, see the ESI (Fig. S1 and Table S1).†

Various Cu sources (Table S2, ESI†) and solvents (Table S3, ESI†) were also examined, and it was found that the system comprising the CuCl/TMEDA/DMN-AZADO catalyst in benzotrifluoride was the most effective for oxidative esterification of **1b** with **1a**. Although the solvent to give the highest **1c** yield was benzotrifluoride, greener solvents⁵⁵ like acetate esters and ketones can also be used for this esterification (Table S4, entries 3–7, ESI†). Moreover, the use of **1a** as the solvent or no use of solvents afforded **1c** in moderate yields (Table S4, entries 8 and 9, ESI†). In particular, when anisole, which is also a recommended green aromatic solvent,⁵⁵ was used for this reaction, the yield of **1c** was equivalent to that in benzotrifluoride (Table S4, entries 1 and 2, ESI†). Therefore, this system can be applied to the efficient synthesis of **1c** in environmentally friendly solvents.

When the reaction time was increased from 10 min to 24 h, the yield of **1c** increased to 93% (Table S5,† entry 1). When oxalic acid or glycolic acid was used as the starting substrate instead of ethylene glycol, the reactions did not proceed at all, confirming that this reaction is not an esterification reaction between carboxylic acids and alcohols (Table S5,† entries 2 and 3). The yields of **1c** decreased drastically (Table S5,† entries 4–6) in the absence of CuCl, TMEDA or DMN-AZADO. Moreover, a significant decrease in the **1c** yield was observed when MS-3A was not added (Table S5,† entry 7). Control experiments on the amount of added H₂O and MS-3A revealed the inhibitory effect of H₂O on the reaction (Table S6†). In the reaction without MS-3A, neither glycolic acid nor oxalic acid was observed, and when the reaction from **1c** was carried out

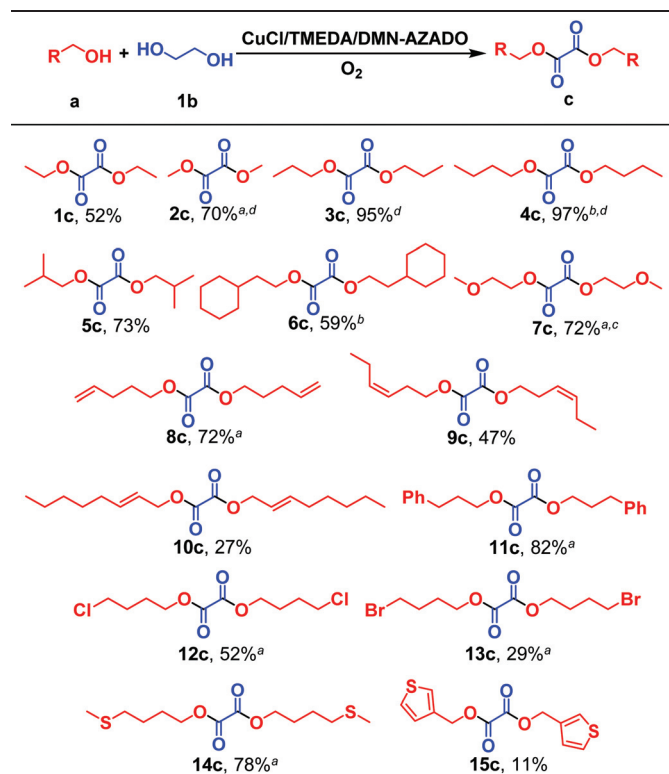
in the presence of H₂O, hydrolysis of **1c** to carboxylic acids did not occur (Scheme S1†). After the reaction without MS-3A, blue droplet formation was observed in the reaction solution, which can be regarded as the H₂O containing Cu species (Fig. S4†). These results indicated that the Cu catalyst was deactivated by H₂O without MS-3A. In addition, under an Ar atmosphere, the oxidative esterification did not proceed, which demonstrates the role of O₂ as the terminal oxidant (Table S5,† entry 8).

Substrate scope for the synthesis of oxalic acid diesters

Under the optimized reaction conditions, we examined the substrate scope for the proposed CuCl/TMEDA/DMN-AZADO-catalyzed oxidative esterification of ethylene glycol with alcohols. As a result, structurally diverse oxalic acid diesters could be synthesized starting from various kinds of primary/secondary alcohols (Tables 3, 4 and Scheme 1). The gas chromatography yields of desired oxalic acid diesters and byproducts (aldehydes/ketones and homo-esterification products) are summarized in Tables S7 and S8, ESI.†

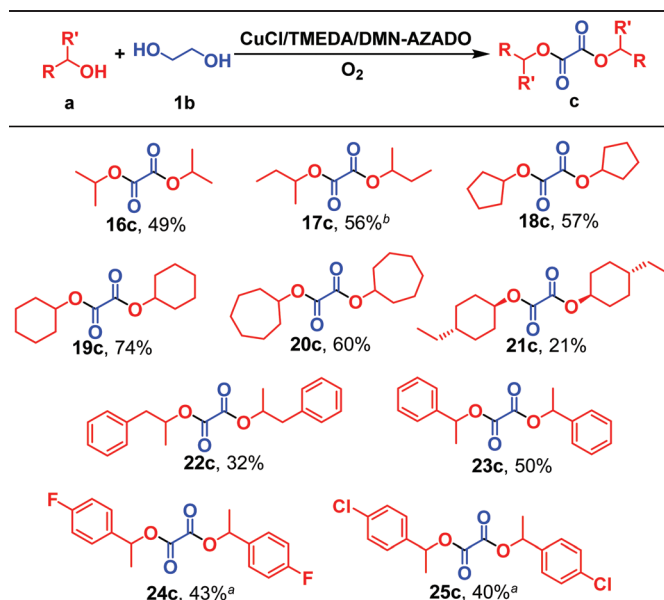
First, we investigated primary alcohols for this oxidative esterification (Table 3). Linear aliphatic alcohols reacted well to give oxalic acid diesters (**1c–4c**). Branched or cyclohexyl ali-

Table 3 Substrate scope for the oxidative esterification of ethylene glycol and primary alcohols



Reaction conditions: **a** (1.1 mmol), **1b** (0.55 mmol), CuCl (0.025 mmol), TMEDA (0.025 mmol), DMN-AZADO (0.015 mmol), benzotrifluoride (1.6 mL), O₂ (balloon), MS-3A (200 mg), room temperature, 24 h, PhCl (0.1 mmol) as an internal standard. Isolated yields are shown. ^a MS-3A (500 mg). ^b 1 h. ^c 96 h. ^d GC yields are shown.

Table 4 Substrate scope for the oxidative esterification of ethylene glycol and secondary alcohols

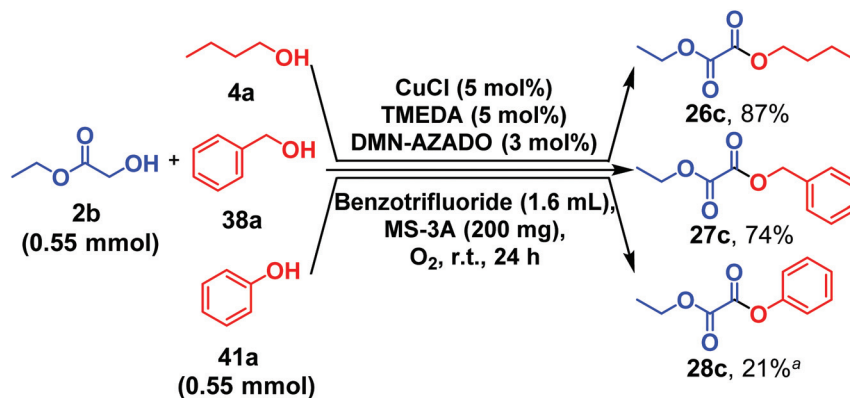


Reaction conditions: **a** (1.1 mmol), **1b** (0.55 mmol), CuCl (0.025 mmol), TMEDA (0.025 mmol), DMN-AZADO (0.015 mmol), *o*-xylene (1.6 mL), O₂ (balloon), MS-3A (500 mg), room temperature, 24 h, PhCl (0.1 mmol) as an internal standard. Isolated yields are shown. ^a **a** (5.5 mmol). ^b GC yield is shown.

phatic alcohols also proved suitable for this esterification (**5c**, **6c**). The reaction proceeded smoothly with an alcohol having an ether structure (**7c**) and with unsaturated alcohols containing C=C double bonds or a phenyl group (**8c–11c**). Ethylene glycol was selectively oxidized in the presence of an allylic alcohol, yielding the desired oxalic acid diester (**10c**). The reaction also proceeded with alcohols having heterofunctional groups such as chloride, bromide, sulfide and thiophene groups (**12c–15c**).

Next, the substrate scope of secondary alcohols was examined using the optimized CuCl/TMEDA/DMN-AZADO catalyst system (Table 4).⁵⁶ The oxidative esterification of simple aliphatic alcohols proceeded (**16c**, **17c**). The corresponding diesters were also obtained using cyclic alcohols as the substrates (**18c–21c**). Surprisingly, the use of alcohols with aryl groups including benzylic alcohol groups gave the desired diesters (**22c–25c**). The reaction also proceeded with alcohols having heterofunctional groups such as fluoride and chloride groups (**24c**, **25c**). Despite the good results obtained for many kinds of oxalic acid diesters, the reaction did not work well with some alcohols, including benzylic alcohols and phenol (Table S9, ESI†).

Finally, we investigated the synthesis of unsymmetrical oxalic acid diesters. When ethylene glycol (**1b**) and two different alcohols were used as the starting substrates, the corresponding unsymmetrical diester was not obtained selectively owing to the competitive addition of each alcohol (Scheme S2, ESI†). Then we evaluated the use of ethyl glycolate



Scheme 1 Synthesis of unsymmetrical oxalic acid diesters. Isolated yields are shown. ^aGC yield determined by using PhCl (0.1 mmol) as an internal standard is shown.

(2b) as the starting substrate for oxidative esterification reactions with various alcohols to synthesize unsymmetrical diesters (Scheme 1). As a result, the corresponding diesters were selectively obtained, not only using a simple alkyl alcohol but also using benzyl alcohol and phenol.

Mechanistic study of the present selective alcohol oxidation by CuCl/TMEDA/DMN-AZADO

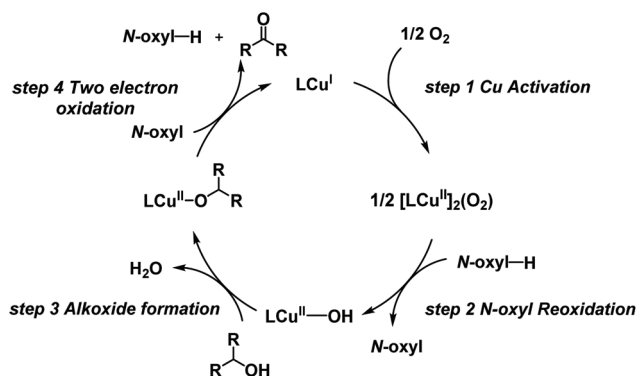
It is known that the Cu/*N*-oxyl-catalyzed alcohol oxidation reaction comprises four steps (Scheme 2): the activation of a Cu(I) species by O₂ to generate a dicopper-oxygen complex (step 1); the oxidation of a hydroxylamine to an *N*-oxyl by the activated Cu(II) with the formation of a Cu(II)–OH species (step 2); Cu(II)–alkoxide formation *via* coordination of an alcohol to the Cu(II)–OH species (step 3); and a two-electron oxidation by the *N*-oxyl and the Cu(II) species to afford the corresponding carbonyl compound (step 4).^{57,58} The proposed total catalytic cycle for the oxidative esterification of alcohols with ethylene glycol to oxalic acid diesters is shown in Scheme S3.[†]⁵⁹

On the basis of this mechanism, the CuCl/TMEDA/DMN-AZADO catalyst successfully oxidized ethylene glycol with high selectivity even in the presence of other alcohols. To rule out the possibility that the present catalytic system could not oxidize primary alcohols such as ethanol, we performed separ-

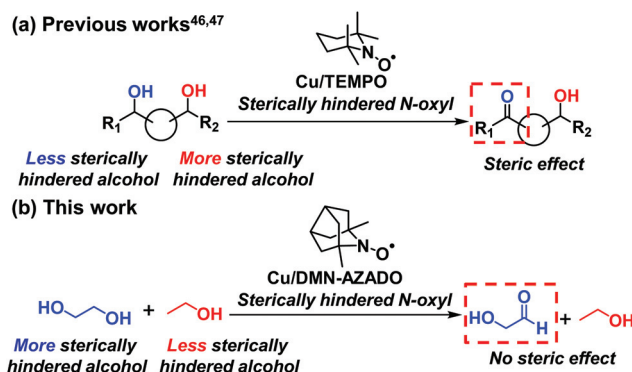
ately the oxidation of ethanol (1a) and ethylene glycol (1b) using the CuCl/TMEDA/DMN-AZADO catalyst. As a result, both alcohols were oxidized well, in contrast with the competitive oxidation of 1a and 1b (Fig. S5, ESI[†]). In other words, the oxidation of ethanol was almost completely suppressed by the presence of ethylene glycol.

In general, the selectivity of the Cu/*N*-oxyl-catalyzed alcohol oxidation is greatly influenced by the steric effect of the alcohol and the *N*-oxyl. This selectivity is determined in the reaction between the Cu(II)–alkoxide and the *N*-oxyl depicted in the step 4 of Scheme 2. For example, when TEMPO, which has a large steric hindrance, is used with Cu(OTf) to oxidize alcohols, primary alcohols are selectively oxidized (Scheme 3a),^{46,47} whereas a Cu(OTf)/ABNO catalyst can oxidize both primary and secondary alcohols.⁵⁰ However, it is unlikely that the selectivity in this study stems from such steric effects (Scheme 3b). As shown in Table 1, when diethyl oxalate (1c) is synthesized from ethanol (1a) and ethylene glycol (1b) with the present Cu/*N*-oxyl system, ethylene glycol, which has a larger steric hindrance than ethanol, is selectively oxidized using any *N*-oxyl other than TEMPO, including DMN-AZADO.

The acidity of alcohols has been reported as another factor that determines the reaction rate of the Cu/*N*-oxyl-catalyzed alcohol oxidation.^{57,60} In previous studies, alcohols with lower



Scheme 2 Reaction mechanism of the Cu/*N*-oxyl-catalyzed alcohol oxidation.



Scheme 3 Effect of the steric hindrance of *N*-oxyls on the selectivity to Cu/*N*-oxyl-catalyzed alcohol oxidation.

pK_a have been proved to be oxidized preferentially in the competitive oxidation of benzylic alcohols owing to fast Cu(II)-alkoxide formation (Scheme 2, step 3) facilitated by electron-withdrawing substituents (Scheme 4a). Accordingly, the relatively higher acidity ($pK_a = 15.1$ in H_2O)⁶¹ of ethylene glycol compared with other alcohols owing to the electron-withdrawing hydroxy group at the β -position can be invoked to explain the high selectivity in this reaction. To evaluate the effect of the alcohol acidities on the present selective oxidation, we conducted competitive alcohol oxidations (**4a** vs. **7a**, **7a** vs. **39a** and **39a** vs. **4a**) using 1-butanol (**4a**, $pK_a = 16.1$ in H_2O),⁶² 2-methoxy ethanol (**7a**, $pK_a = 14.8$ in H_2O)⁶¹ and 3,3,3-trifluoro-1-propanol (**39a**, $pK_a = 12.7$ in a mixture of H_2O and ethanol (1:1)),⁶³ in the presence of the CuCl/TMEDA/DMN-AZADO catalyst. The results revealed the absence of reaction rate dependence on the alcohol acidities and that the oxidation of all the three alcohols proceeded even under competitive oxidation conditions (Fig. S6 and Scheme S4, ESI†). Furthermore, 2-methoxyethanol, which has a methoxy group at the β -position like ethylene glycol, has a lower pK_a than ethylene glycol;⁶¹ however, ethylene glycol was oxidized preferentially over 2-methoxyethanol in the presence of CuCl/TMEDA/DMN-AZADO (Table 3, **7c**). These results indicate that the acidity of alcohols is not the dominant factor determining the selectivity in this reaction (Scheme 4b).

To elucidate the mechanism for this unique selectivity, we first performed competitive oxidation reactions of various alcohols including diols, hydroxyketones and hydroxyesters (**b**) and ethanol (**1a**) in the presence of the CuCl/TMEDA/DMN-AZADO catalyst and then we calculated the excess ratio of products (**f**) obtained by oxidation of **b** to all of the observed products in the reaction (Fig. 3 and Scheme S5, ESI†). As mentioned above, ethylene glycol (**1b**) was preferentially oxidized over ethanol (**1a**), affording a high excess ratio of **f**. In contrast, the selectivity for the oxidation of *n*-propanol (**4b**) was quite low, suggesting the importance of a diol structure like that of ethylene glycol for the selective oxidation. Next, in the case of 1,3-propanediol (**5b**), which is a primary alcohol with the other hydroxy group at the γ -position, the selectivity to **5b** oxidation remained low. Meanwhile, the use of 1,2-propanediol (**6b**), which has a hydroxy group at the β -position, showed comparatively high selectivity. Furthermore, compounds with a carbonyl group at the β -position of the hydroxy group (**7b**, **2b**) were oxidized

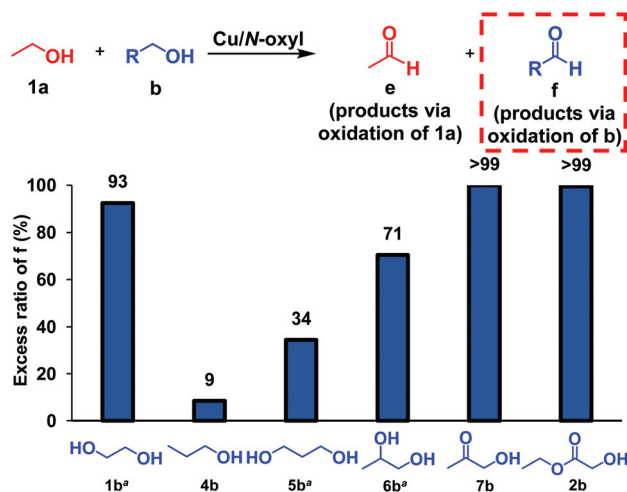
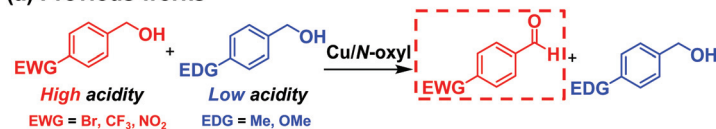


Fig. 3 Selectivity of alcohol (**b**) oxidation to the corresponding aldehyde-derived products in the presence of ethanol (**1a**). Reaction conditions: **1a** (0.55 mmol), **b** (0.55 mmol), CuCl (0.025 mmol), TMEDA (0.025 mmol), DMN-AZADO (0.015 mmol), THF (1.6 mL), O_2 (balloon), MS-3A (200 mg), room temperature, 1 h, PhCl (0.1 mmol) as an internal standard. Excess ratio of **f** = $(f - e) \text{ (mmol)} \times 100 / (f + e) \text{ (mmol)}$. See Scheme S5, ESI† for the detailed calculation of the excess ratio of **f**. ^a**1a** (1.1 mmol).

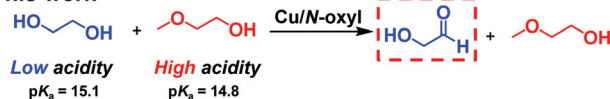
highly selectively. These results indicate that the presence of a heterofunctional group such as a hydroxy or carbonyl group at the β -position of the alcoholic hydroxy group has a significant effect on the selectivity of the present alcohol oxidation.

To gain a mechanistic insight into the effects of these functional groups at the β -positions in alcohols, solutions containing CuCl/TMEDA and alcohols were characterized by CSI-MS (positive mode, Fig. 4). A CSI-MS spectrum of the mixture of CH_3CN solvent, CuCl, TMEDA and ethanol (**1a**) showed sets of signals at $m/z = 267.20$ and $m/z = 447.23$ that can be attributed to $[Cu(II)TMEDA(CH_3CH_2O)(CH_3CHNH)]^+$ and $[Cu(II)_2Cl_2(OH)(TMEDA)_2]^+$ (Fig. 4a).⁶⁴ In spite of an excess amount of ethanol, the dinuclear Cu(II)/TMEDA complex remained in the solution, indicating that the coordination of ethanol to the Cu(II) species is comparatively difficult. Meanwhile, the CSI-MS spectrum of the mixture of CH_3CN solvent, CuCl, TMEDA and ethyl glycolate (**2b**) exhibited a set of strong signals at $m/z = 282.18$ that can be attributed to $[Cu(II)TMEDA(C_4H_7O_3)]^+$,

(a) Previous works^{57,60}



(b) This work



Scheme 4 Effects of acidity on the selectivity to Cu/N-oxyl-catalyzed alcohol oxidation.

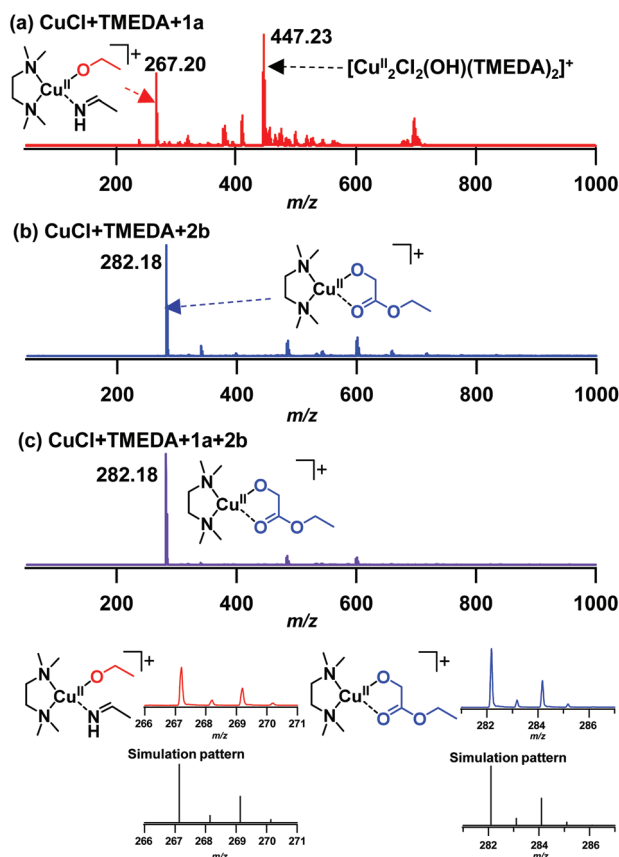


Fig. 4 CSI-MS spectra of Cu(II)/TMEDA/alcohols: (a) **1a**; (b) **2a**; (c) **1a** and **2a**. Measurement conditions: positive-ion CSI-MS of a CH₃CN solution (1.6 mL) containing **1a** and/or **2a** (1.1 mmol), CuCl (0.025 mmol), TMEDA (0.025 mmol), and MS-3A (200 mg) stirred for 1 h at room temperature under an O₂ atmosphere. Inset: the observed signal at $m/z = 267.20$ and the simulated pattern for [Cu(II)TMEDA(CH₃CH₂O)(CH₃CHNH)]⁺, and the observed signal at $m/z = 282.18$ and the simulated pattern for [Cu(II)TMEDA(C₄H₇O₃)]⁺.

without the aforementioned dinuclear Cu(II)/TMEDA complex (Fig. 4b). This result revealed the strong bidentate coordination to Cu(II). In addition, the CSI-MS spectrum of a CH₃CN solution containing CuCl, TMEDA, **1a** and **2b** showed the same signals for the bidentate ethyl glycolate–Cu(II) complex as those obtained using **2b** alone (Fig. 4c). Taken together, these results suggest that alcohols having a hydroxy or a carbonyl group at the β -positions preferentially coordinated to Cu(II) in a bidentate fashion, which explains their selective oxidation in this catalytic system.

To clarify the mechanism of this oxidation in more detail, CuCl/TMEDA/DMN-AZADO-catalyzed oxidations of ethanol (**1a**) and ethylene glycol (**1b**) were examined by density functional theory (DFT) computational studies referring to the previous report⁵⁸ (Fig. 5). It was confirmed that there was almost no energy difference between ethanol and ethylene glycol in the two-electron/one-proton transfer from alcohols to Cu/DMN-AZADO to produce the corresponding aldehydes (6 $\rightarrow$ 8). In contrast, in the process of coordination to Cu(II) species as alkoxides (1 $\rightarrow$ 4), the complex formed by ethylene glycol was considerably more stable than that by ethanol owing to its bidentate coordination. Furthermore, the activation energy in the proton transfer process of ethylene glycol (2 $\rightarrow$ 3-TS) was smaller than that of ethanol. Since the DFT calculation revealed that the reaction of ethylene glycol was less favoured than that of ethanol if the reaction proceeded *via* monodentate coordination of ethylene glycol (Fig. S7, ESI[†]), it seems reasonable to conclude that the bidentate coordination through the hydroxy group at the β -position occurred. These results indicate that this selective oxidation of ethylene glycol-derived alcohols/hemiacetals is *via* the kinetically and thermodynamically favourable bidentate coordination of the alcohols having hydroxy or carbonyl groups at the β -positions to the Cu(II) catalyst.

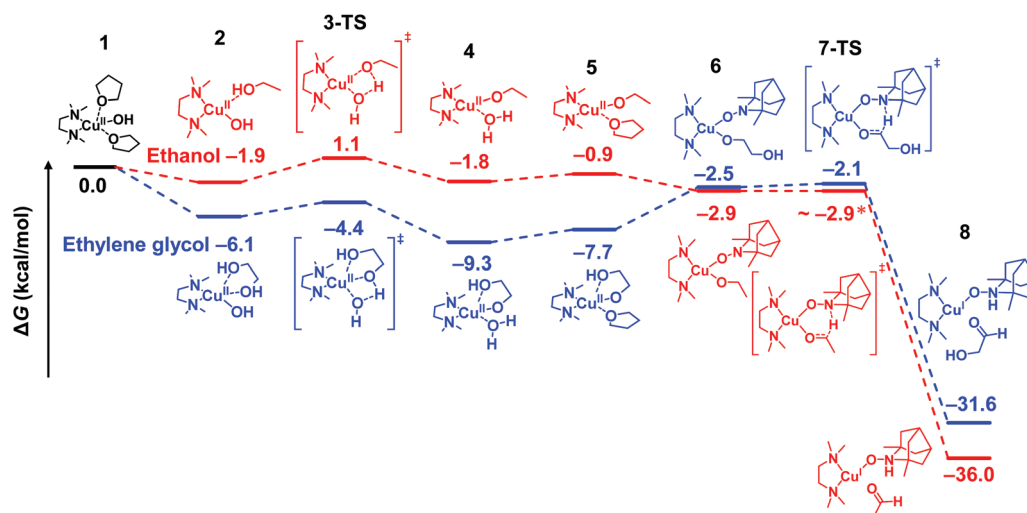


Fig. 5 Density functional theory calculations in ethanol oxidation and ethylene glycol oxidation catalyzed by CuCl/TMEDA/DMN-AZADO. *See the ESI[†] for details.

In addition, we compared the potential energy diagrams of several catalysts for the oxidation of ethylene glycol or ethyl glyoxylate hemiacetal by DFT calculations (Fig. S8–S10, ESI†), which are consistent with the above-mentioned experimental data.⁶⁵ Overall, it can be concluded that this efficient oxidation *via* a stable chelate complex is enabled by a quick two-electron/one-proton transfer from ethylene glycol-derived alcohols/hemiacetals to Cu/DMN-AZADO. Therefore, the designed CuCl/TMEDA/DMN-AZADO system prevents catalyst deactivation by the chelation effect of ethylene glycol-derived alcohols/hemiacetals; instead, it enables the highly selective and efficient oxidation of the alcohols.

Conclusion

We have developed a novel method for the selective intermolecular oxidation of aliphatic primary alcohols enabling the synthesis of oxalic acid diesters *via* oxidative esterification from alcohols and ethylene glycol using the CuCl/TMEDA/DMN-AZADO catalyst. The reaction is environmentally friendly, proceeding under mild reaction conditions using O₂ as the terminal oxidant, and surprisingly, it only requires a stoichiometric amount of alcohols and ethylene glycol. This catalytic system was applicable to the synthesis of various oxalic acid diesters from structurally diverse primary and secondary alcohols, thus proving its versatility. The unique effect of DMN-AZADO, which was essential for the high catalytic performance, was suggested to be the prolonged lifetime stemming from its appropriate steric hindrance. Thorough experiments and DFT calculations revealed that this unusual high selectivity is mainly due to the presence of a heterofunctional group at the β -position of the alcohols because it enables the bidentate coordination to the Cu(II) species, forming kinetically and thermodynamically favourable Cu(II) complex intermediates. In addition, the present CuCl/TMEDA/DMN-AZADO catalyst showed sufficiently high performance for oxidation from the stable Cu(II) complex *via* an efficient two-electron/one-proton transfer and the TMEDA-promoted reoxidation. We believe that the catalytic system developed in this study will be utilized for the green synthesis of oxalic acid derivatives in various fields and will lead to new concepts of selective molecular transformations enabled by highly active catalysts that overcome deactivation derived from strong coordination.

Conflicts of interest

There are no conflicts to declare.

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